

How to Simulate a Galaxy

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1 Introduction

This is my exploration into simulatin a galaxy using Julia. My goal is to build up a simulation using more and more advanced features as I add to it. This will allow me to learn about the process of simulating a galaxy, as well as best practices for high performance Julia code.

I will start with:

- Basic particle type
- Direct summation of forces
- Runge-Kutta and Symplectic integrators with fixed and/or variable timesteps

Eventually I will add things like:

- Barnes-Hut algorithm for force summation
- Fast Multipole Method for force summation
- Smoothed Particle Hydrodynamics
- Mesh-grid fluid-like simulations

Finally/during the development I will learn how to make these systems parallel. This could include, but is not limited to, multi-threading and GPU acceleration.

1.1 Setting up the package

```
julia --eval 'using Pkg; Pkg.generate("GalaxySim")'

cd GalaxySim
julia --eval 'using Pkg; Pkg.activate("."); Pkg.instantiate()'
```

```
cd GalaxySim
touch main.jl
echo using GalaxySim >> main.jl
```

```
git init .
git add Project.toml
git add main.jl
git add src/
git commit -m 'Initial Commit'
```

1.2 TODO Configure startup.jl

Remember to activate the environment!

- Add this to startup.jl

2 Particle Types

2.1 Basic Particle

A basic particle needs to hold a few key pieces of information.

1. Its mass, to determine how it effects other particles
2. Its position, so we know where it is relative to the other particles
3. Its velocity, so we know how it moves throughout the galaxy.

I in previous projects have been very fond of the Dynamic Form of ODEs as described by [Landau, Páez & BordeianuLandau et al.2024].

using GalaxySim

2.2 Smoothed Particle Hydrodynamics Particle

3 Summation of Forces

3.1 Direct Summation

3.2 Barnes-Hut Algorithm

3.3 Fast Multipole Method

4 Integration of Forces

4.1 Euler Method

4.2 Symplectic Methods

5 Fluid Dynamics

5.1 Smoothed Particle Hydrodynamics

$$\rho(\mathbf{r}) = \sum_{b=1}^{N_{\text{neigh}}} m_b W(\mathbf{r} - \mathbf{r}_b, h_b)$$

5.2 Grid based methods

References

[Landau, Páez & BordeianuLandau et al.2024] Landau R., Páez M., Bordeianu C., 2024, Computational Physics: Problem Solving with Python, 4 edn. Wiley-VCH, Weinheim, Germany